

Basic Principles of Medicine 1

Module: Foundations To Medicine| Lecture 35

Histology of Muscle Tissue - Flipped Class Pre-Lecture Package (1st Recording – General Overview)

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Recommended Reading

Histology – A Text and Atlas

Pawlina 9th Edition

Chapter 11: Muscle Tissue Pg. 344-387

- <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com.sgu.idm.oclc.org/book.aspx?bookid=3290>



Objectives

List the functions of muscle tissue.	SOM.MK.I.BPM1.1.FTM.3.HCB.o887
Describe the structure of thick myofilaments in striated muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.o888
Identify the proteins associated with the thick myofilament in striated muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.o889
Give the function(s) of the proteins associated with the thick myofilament in striated muscles	SOM.MK.I.BPM1.1.FTM.3.HCB.o890
Describe the structure of thin myofilaments in striated muscles	SOM.MK.I.BPM1.1.FTM.3.HCB.o891
Identify the proteins associated with the thin myofilament in striated muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.o892
Give the function(s) of the proteins associated with the thin myofilament in striated muscles	SOM.MK.I.BPM1.1.FTM.3.HCB.o893
Give the function of the sarcomere.	SOM.MK.I.BPM1.1.FTM.3.HCB.o894
Describe the microscopic structure of the sarcomere.	SOM.MK.I.BPM1.1.FTM.3.HCB.o895
Describe the sliding filament model of muscle contraction.	SOM.MK.I.BPM1.1.FTM.3.HCB.o896
Describe briefly the events of the actomyosin cross-bridge cycle.	SOM.MK.I.BPM1.1.FTM.3.HCB.o897
List the 2 types of striated muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.o898
Describe the contractile apparatus of striated muscle fibers.	SOM.MK.I.BPM1.1.FTM.3.HCB.o899
State the functions of the accessory proteins typically associated with striated muscle cells.	SOM.MK.I.BPM1.1.FTM.3.HCB.o900
Give the function(s) of skeletal muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.o901



Objectives

State the localization of skeletal muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0902
Describe the microscopic structure of skeletal muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0903
Describe the general organization of skeletal muscle including connective tissue, muscle fascicles, muscle fibers, myofibrils and myofilaments.	SOM.MK.I.BPM1.1.FTM.3.HCB.0904
Describe the structure, function and localization of the T-tubules and the sarcoplasmic reticulum in skeletal muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0905
Describe the three types of skeletal muscle fibers.	SOM.MK.I.BPM1.1.FTM.3.HCB.0906
Discuss the regenerative capability of skeletal muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0907
Describe the ultrastructural defects in muscular dystrophies including Duchenne muscular dystrophy.	SOM.MK.I.BPM1.1.FTM.3.HCB.0908
Give the function of the muscle spindle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0909
Describe the histological structure of the muscle spindle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0910
Give the function of the Golgi tendon organs.	SOM.MK.I.BPM1.1.FTM.3.HCB.0911
Describe the histological structure of the Golgi tendon organs.	SOM.MK.I.BPM1.1.FTM.3.HCB.0912
Give the function(s) of cardiac muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0913
State the localization of cardiac muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0914
Describe the microscopic structure of cardiac muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0915



Objectives

Describe the general organization of cardiac muscle, including the structure and function of intercalated discs.	SOM.MK.I.BPM1.1.FTM.3.HCB.0916
Give the function(s) of Purkinje fibers	SOM.MK.I.BPM1.1.FTM.3.HCB.0917
Describe the microscopic structure Purkinje fibers.	SOM.MK.I.BPM1.1.FTM.3.HCB.0918
State the localization of Purkinje fibers.	SOM.MK.I.BPM1.1.FTM.3.HCB.0919
Describe the structure, function and localization of the T-tubules and the sarcoplasmic reticulum in cardiac muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0920
Discuss the regenerative capability of cardiac muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0921
Give the function(s) of smooth muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0922
State the localization of smooth muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0923
Describe the microscopic structure of smooth muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0924
Describe the microscopic structure of the thick myofilaments in smooth muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.0925
Identify the proteins associated with the thick myofilament in smooth muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.0926
Give the function(s) of the proteins typically associated with the thick myofilament in smooth muscles	SOM.MK.I.BPM1.1.FTM.3.HCB.0927

Objectives

Describe the microscopic structure of thin myofilaments in smooth muscles	SOM.MK.I.BPM1.1.FTM.3.HCB.0928
Identify the proteins associated with the thin myofilament in smooth muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.0929
Give the function(s) of the proteins typically associated with the thin myofilament in smooth muscles.	SOM.MK.I.BPM1.1.FTM.3.HCB.0930
Describe the contractile apparatus of smooth muscle cells.	SOM.MK.I.BPM1.1.FTM.3.HCB.0931
Describe the regenerative capability of smooth muscle.	SOM.MK.I.BPM1.1.FTM.3.HCB.0932
State the function(s) of the accessory proteins usually associated with smooth muscle cells.	SOM.MK.I.BPM1.1.FTM.3.HCB.0933

Tissue Overview

- Tissue - the organization of individual cells into groups to perform a specific function(s)

4 Basic Tissue Types

1. Epithelial tissue

- Covers body surfaces, lines body cavities, and forms glands
- Originates from all 3 germ cell layers
- a. Ectoderm- e.g., epidermis (skin), epithelium of- oral cavity, eyes, ears
- b. Mesoderm – epithelium of blood vessels (endothelium) and lining of internal cavities (mesothelium) (e.g., peritoneal, pleural cavities)
- c. Endoderm – e.g., epithelium lining respiratory system, digestive tract, liver, gallbladder, pancreas, thyroid, parathyroid glands

2. Connective tissue (CT)

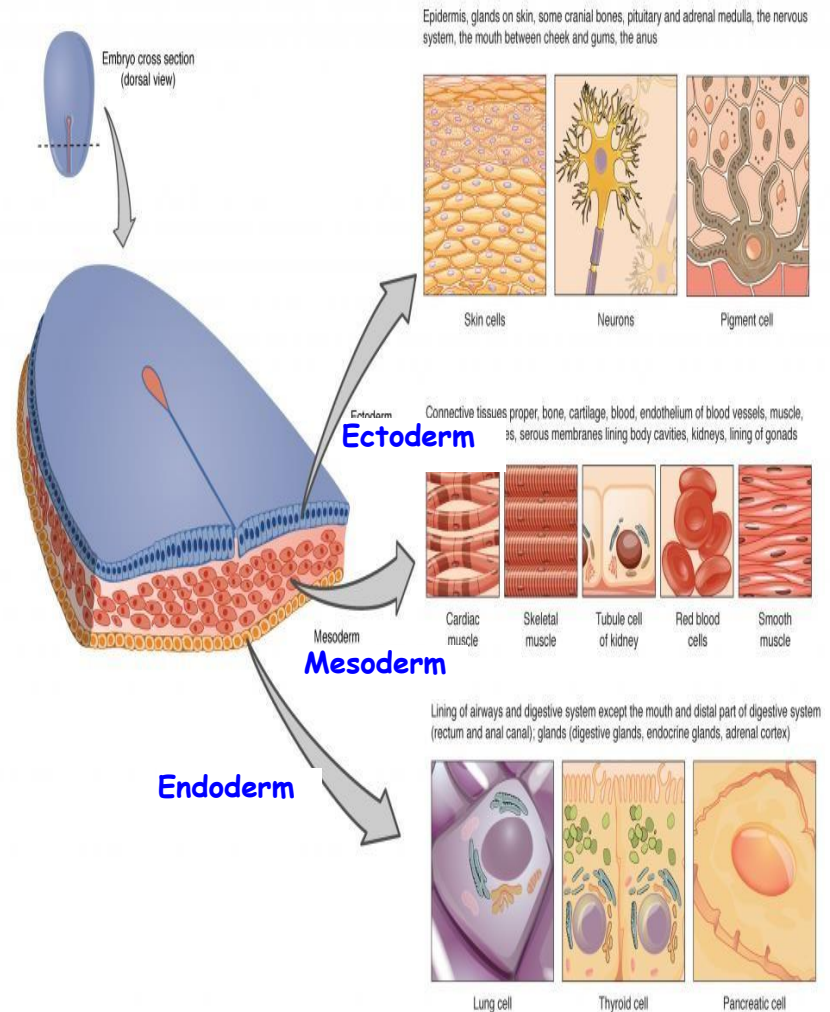
- Underlies or supports the other three basic tissues structurally & functionally
- Embryonic CT derived from mesodermal germ cell layer

3. Muscle tissue

- Consists of contractile cells and responsible for movement
- Derived from the mesoderm

4. Nerve tissue

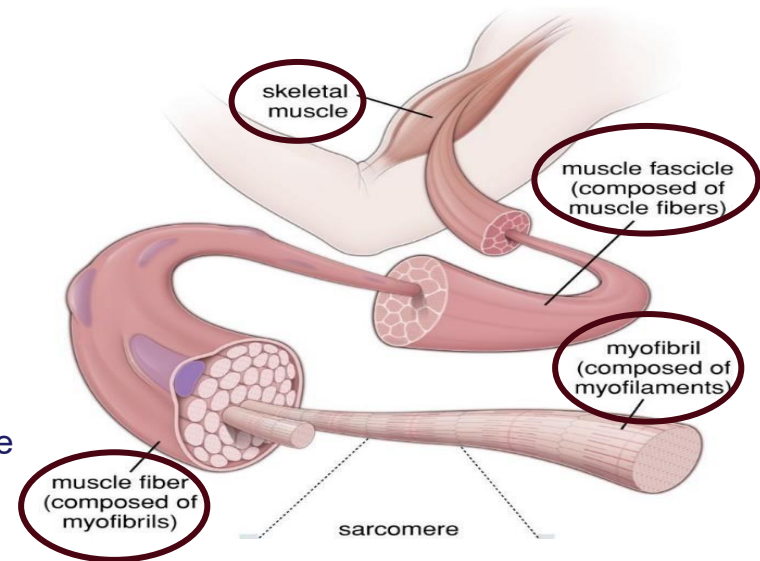
- Receives, transmits and integrates information from outside & inside the body to control activities of the body
- Mainly originates from the neuroectoderm



General Terms associated with Muscle Tissue

Myo

- Myofilaments – contractile proteins
 - e.g. actin and myosin
- Myofibril – bundles of myofilament
 - Structural and functional subunit of the cell
- Myofiber (muscle fiber/muscle cell) - composed of bundles of myofibrils
 - Grouped together in **fascicles or bundles** to make the gross muscle structure
 - Also called myocyte in cardiac tissue



Sarco

- Sarcolemma – plasma membrane
- Sarcoplasm – cytoplasm
- Sarcoplasmic reticulum – smooth endoplasmic reticulum
- Sarcomere – the specific organization of the contractile proteins of the myofibril
 - Observed in all types of striated muscle types (skeletal and cardiac)

Function of Muscle Tissue

- Movement of body parts
- *Changes in* volume and shape of internal organs and vessels
- When summed, these can be brought down to contraction
- Therefore, the most basic function of muscle tissue = **contraction**

What Makes a Muscle Contract?

Myofilaments

- These are **contractile proteins/elements**
 - Occupy the bulk of the sarcoplasm
 - Aligned close to each other and interact with each other for contraction
 - This allows for morphological classification based on appearance under light microscope
 - Smooth muscle
 - Striated muscle
- 2 types of myofilaments
 - Thin filaments
 - Composed mainly of F-actin
 - Thick filaments
 - Composed mainly of **myosin II** motor protein (in striated muscles) and **smooth muscle myosin (SMM)** in smooth muscle cells
 - The thick and thin filaments aggregate together to form a **myofibril**
- Both types of filaments have associated and accessory proteins

Myofilaments and Their Associated Proteins

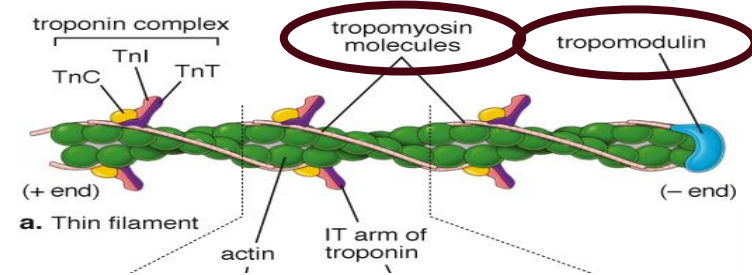
– Straited Muscles

Thin filaments

- Composed of the actin subunits (F-actin)
- Has binding site for myosin
 - Site is covered by **tropomyosin** in resting state
 - Each tropomyosin is associated with one (1) **troponin** complex
- Polarized
 - Plus (+) end is bound to the Z line by **α -actinin** and **nebulin**
 - Minus (-) end extends toward the M line
 - Protected by **tropomodulin** (actin capping protein)

Thin Filament Associated Proteins

- Also considered regulatory proteins
 - Tropomyosin
 - Covers the myosin binding site on the actin filament in the resting muscle state
 - Tropomodulin
 - capping protein which maintains & regulates length of the actin filaments
 - Troponin complex
 - Troponin C (TnC) – binds calcium
 - Troponin T (TnT) – binds tropomyosin and anchors troponin complex
 - Troponin I (TnI) – inhibits actin-myosin interaction
 - Nebulin
 - Stabilizes actin filaments at the Z-line
 - Alpha actinin
 - Bundles actin filaments in parallel array
 - Anchors actin filaments at the Z-line
 - Dystrophin
 - Links the thin filament to laminin



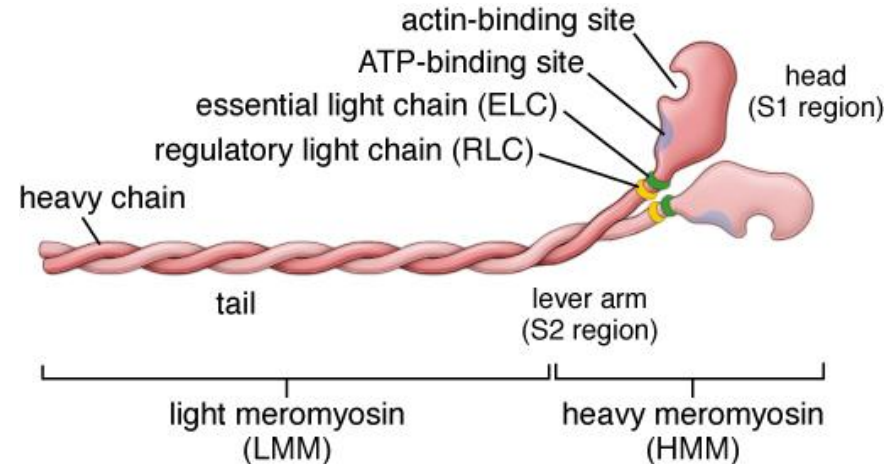
Myofilaments and Their Associated Proteins – Striated Muscles

Myosin II/thick filament

- Binds to actin subunits in **striated** muscles to produce movement
- Has a head and a tail region
- **Head (S1) Region**
 - Connected to the tail region via the lever arm (S2 region)
 - 2 globular heads
 - Project at approximate right angles from the myosin molecule
 - Has 2 specific binding sites
 - One for ATP (with **myosin ATPase** activity)
 - One for actin
- **Tail region**
 - Opposite to the head
 - Aggregated to form a **bipolar thick myosin filament** (in striated muscles only)
 - “Bare zone” in the middle
 - has no globular heads
 - H band of the sarcomere
 - Connected to each other at this zone by a family of **M line** proteins

Thick Filament Associated Proteins:

- M-line proteins
 - Attaches myosin molecules to each other and to titin e.g. creatine phosphatase
- Myosin binding protein C (MyBP-C)
 - Stabilize thick filaments



a. Nucleation of thick filament assembly



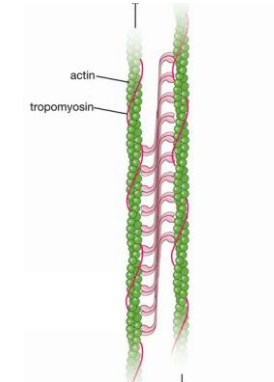
bipolar thick filament

a

Myofilaments and Their Associated Proteins – Smooth Muscles

Thin filaments

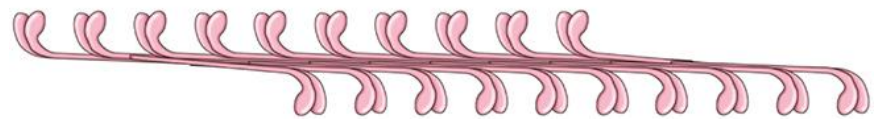
- Double strand of actin subunits
 - Binds to smooth muscle myosin
- **Thin filament Associated Proteins**
 - **Tropomyosin**
 - Smooth muscle isoform
 - Not associated with troponin like is striated muscles
 - Regulated by the phosphorylation of the myosin heads
 - **Caldesmon**
 - Blocks the myosin-binding site on the actin filament
 - **Calponin**
 - Blocks the myosin-binding site on the actin filament



Myofilaments and Their Associated Proteins – Smooth Muscles

Thick Filament

- Composed of myosin subunits
 - Isoform - smooth muscle myosin (**SMM**)
 - SMM is slightly different from the isoform found in striated muscles
 - Like striated muscle isoform, consists of 2 heads and a tail
 - Has the **side-polar arrangement** instead of the bipolar arrangement found in striated muscles
 - SMM molecules are oriented in one direction on one side and the opposite direction on the other side of the filament
 - This arrangement has no central bare zone



side-polar thick filament

b

Thick Filament Associated Proteins

- Myosin light chain kinase (MLCK)
 - Phosphorylates myosin, enabling it to bind to actin



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Histology of Muscle Tissue - Flipped Class Pre-Lecture Package (2nd Recording)

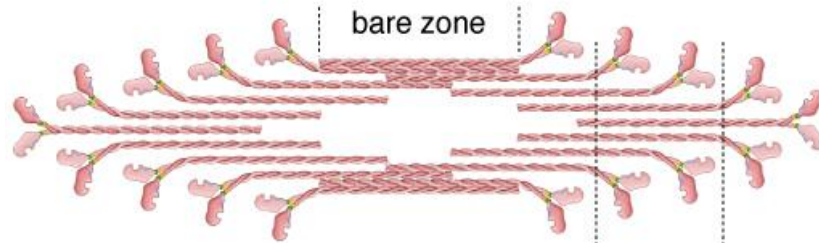
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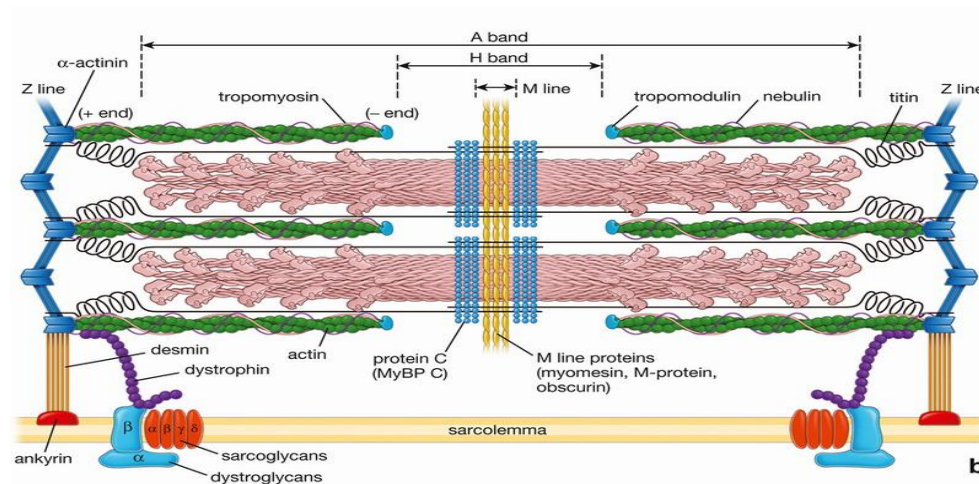
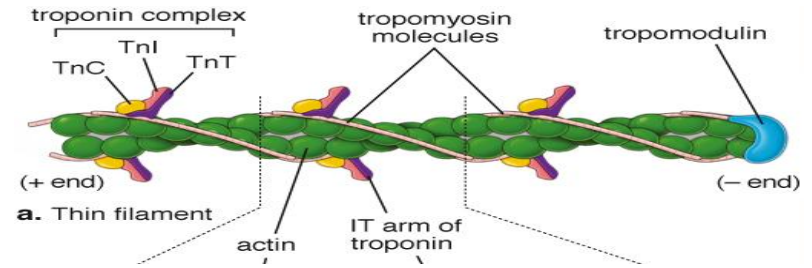
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Myofilaments and the Sarcomere



b. Assembly of myosin in a bipolar thick filament

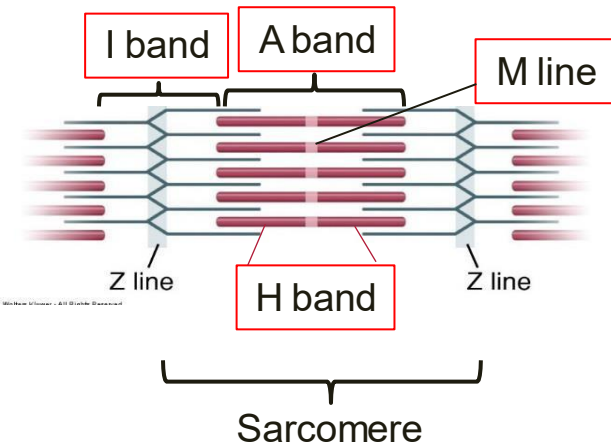


Sarcomere and Striated Muscles

- Specific arrangement of myofilaments creates dark and light bands which accounts for the cross-striation characteristics of **all** striated muscles
 - egs. skeletal and cardiac muscle
- This specific arrangement/organization forms a **sarcomere**

Sarcomere

- Basic contractile unit of striated muscles
- Contains the contractile proteins (myofilaments)
- Defined as the segment of the myofibril between two (2) adjacent Z-lines/discs
- Thick filaments (myosin II) occupy more of the center of the sarcomere
 - Forms the A-band
 - Bisected by the H-band
 - Bisected by the M-line
- Thin filaments (F-actin) are attached to the Z-line
 - Extends into the A-band to the edge of the H-band
- I-band
 - Parts of adjacent sarcomere on either side of the Z-line
 - Contains only thin (F-actin) filaments



Muscle Contraction

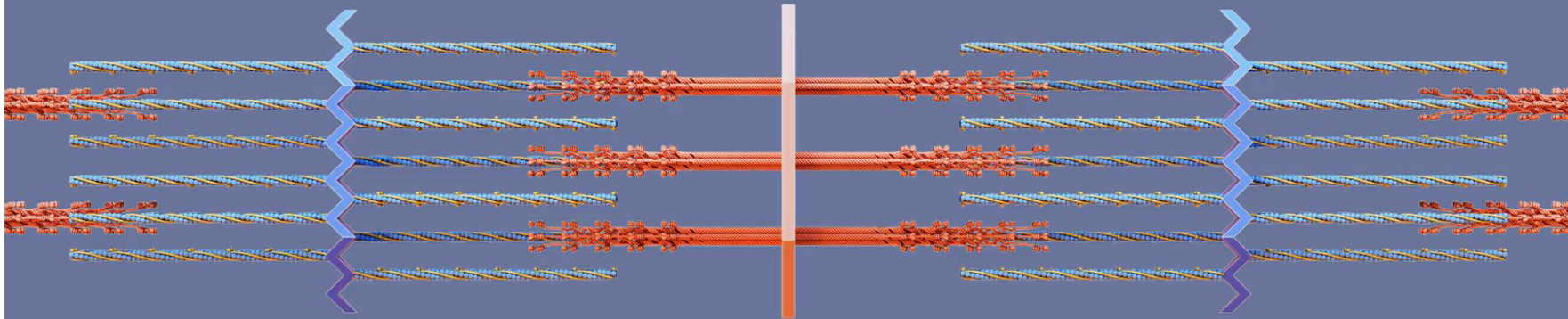
- Muscle contraction is a result of interaction between thick (myosin II) and thin (F-actin) filaments
- In skeletal muscles, many fibers contracting together results in gross movement of that particular muscle group
- **Sliding Filament Hypothesis of Huxley**
 - Sarcomere shortens and becomes thicker, but the myofilaments remain the same length
 - Myofilaments slide past each other
 - Increase the amount of overlap helps them to maintain their lengths
 - Sliding action results from repeated “make and break” attachments between the heads of the myosin molecules and neighboring actin filaments – actomyosin crossbridge
 - A band remains constant
 - I band and H band both decreases in size
 - Z lines are drawn closer to the ends of the A bands
- **Stages of the Contraction Cycle**
- Attachment, release, bending, force generation and reattachment

Think Slide

- Try to imagine:
 - What happens when muscle cells contract?
 - What happens to the I-band, A-band, H-band, M-line?
 - What happens to the myofilaments?
- Use the animation on the following slide to assist you with your response(s)
- **You may want to read up on the Actomyosin Cross-Bridge Cycle from the Pawlina, 9th edition, Chapter 11, section titled The Actomyosin Cross-Bridge Cycle**
- <https://sgu.idm.oclc.org/login?url=https://premiumbasicsciences-lwwhealthlibrary-com.sgu.idm.oclc.org/book.aspx?bookid=3290>

Skeletal Muscle Contraction

Sarcomere



Contractile Apparatus for Striated Muscles

Components

- **Myofilaments**
 - Contractile proteins – F-actin, myosin II
- **Sarcoplasmic reticulum**
 - Well-developed smooth endoplasmic reticulum
 - Membranous compartment of flattened sacs
 - Forms a tubular network around the myofilaments
 - Functions in the rapid delivery and removal of calcium ions from the cytoplasm
 - Has dilated ends called **terminal cisternae**
 - Houses calcium ions
 - Communicates directly with the T-tubules
 - Location varies based on the type of striated muscle
- **Transverse tubules (T-tubules)**
 - Invaginations of the sarcolemma
 - Location varies based on the type of striated muscle
- **Calcium ions (Ca^{2+})**

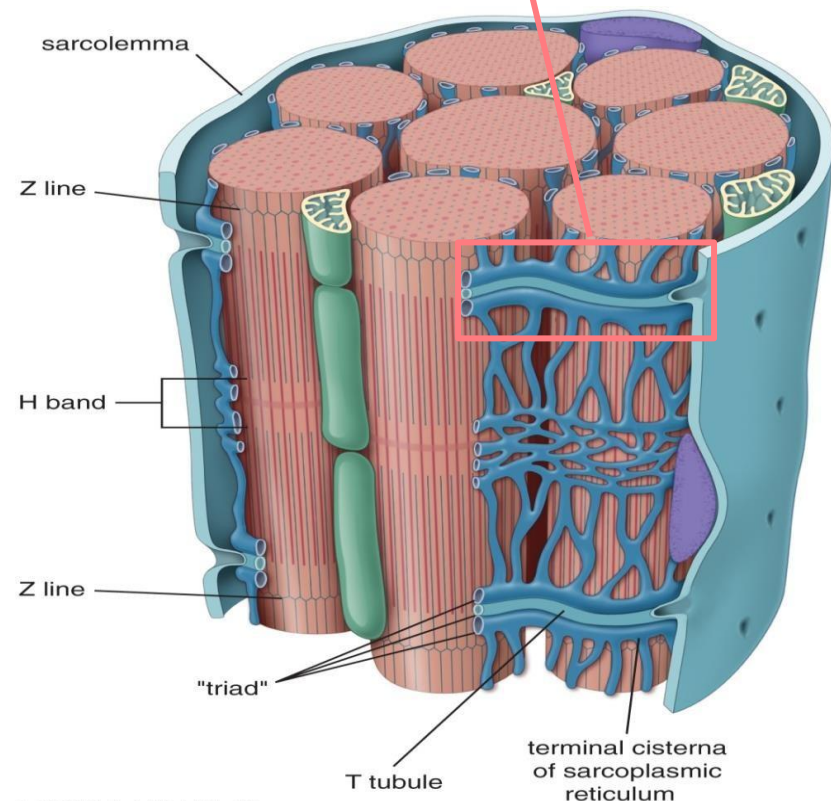
Localization of Contractile Apparatus: Skeletal Muscle

- **Sarcoplasmic reticulum (SER)**
 - Extends from one A-I junction to the next A-I junction within the same sarcomere
- **Terminal cisternae**
 - Enlargements at the A-I junctions that houses calcium ions
- **Transverse tubules (T-tubules)**
 - Invaginations of the sarcolemma at the A-I junction
- **Triad**
 - Found **at the A-I junction**
 - 1 T-tubule and 2 terminal cisternae
 - **Helps to deliver calcium to the cell's cytoplasm**
- Mitochondria scattered between the myofibrils

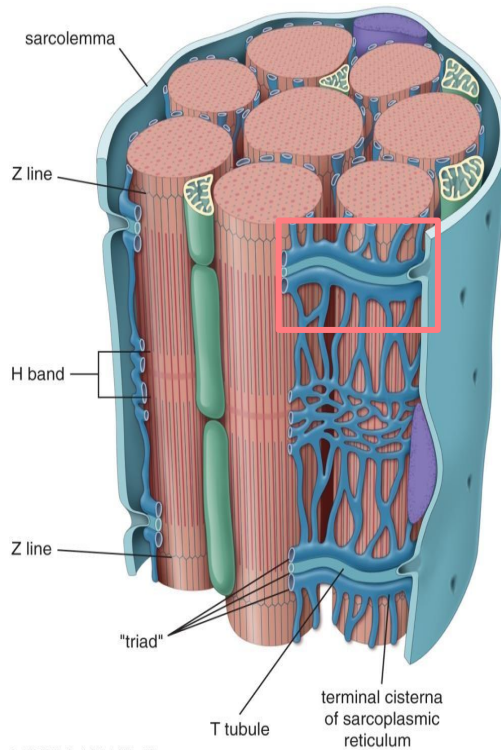
How does it work?

Depolarization of T-tubule membrane triggers release of Ca^{2+} from terminal cisterna & initiates muscle contraction cycle

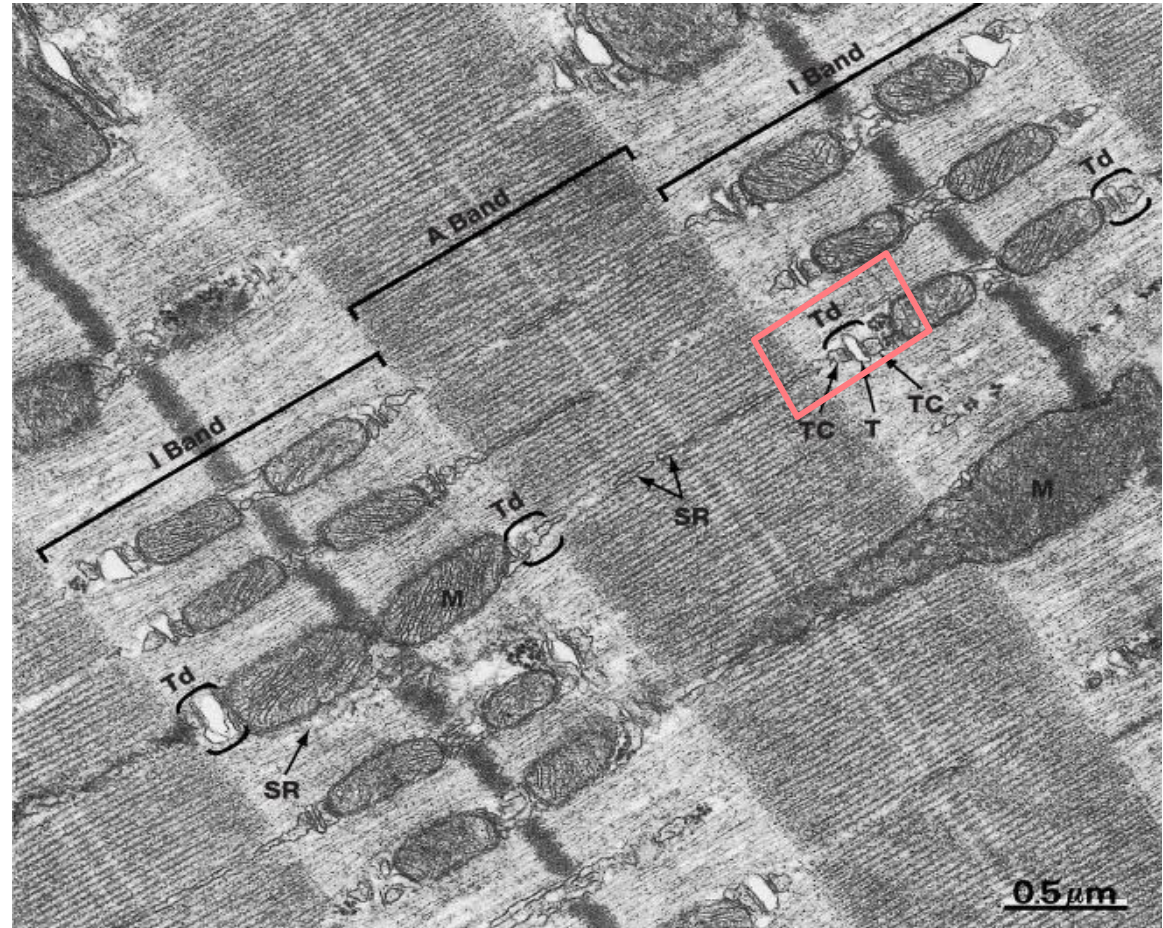
Triad (Td) : 1 T tubule + 2 terminal cisternae at the A-I junction



Localization of Contractile Apparatus: Skeletal Muscle



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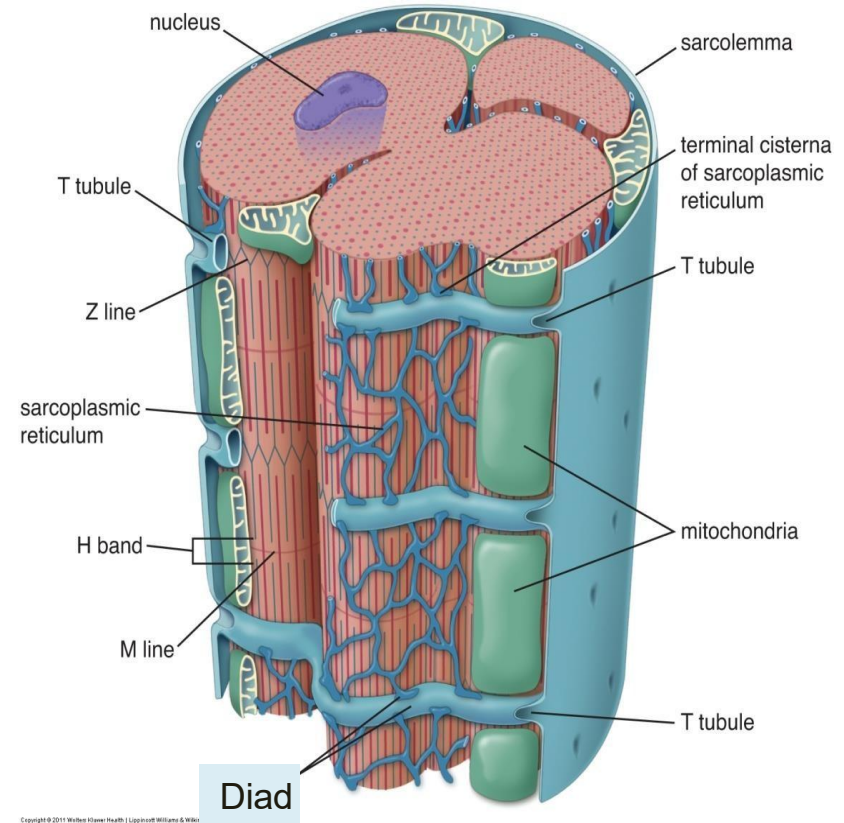
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Localization of Contractile Apparatus: Cardiac Muscle

- **Sarcoplasmic reticulum**
 - Single network along the sarcomere
 - Extends from Z line to Z line
 - Less developed than in skeletal muscle
- **Terminal cisterna**
 - Contain Ca^{2+} release channels
 - Release Ca^{2+} into sarcoplasm
- **Transverse tubules (T-tubules)**
 - Located at the Z line (*compare this to skeletal muscle*)
- **Diad**
 - Complex of one (1) T-tubule and one (1) adjacent terminal cisternae at the **Z-line**
 - Helps to deliver calcium to the cell's cytoplasm

How does it work?

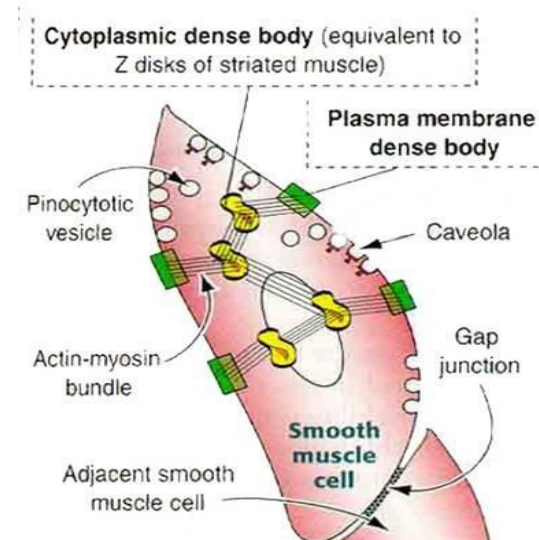
Depolarization of T-tubule membrane triggers release of Ca^{2+} from terminal cisterna & initiates muscle contraction cycle



Contractile Apparatus of Smooth Muscle

Components:

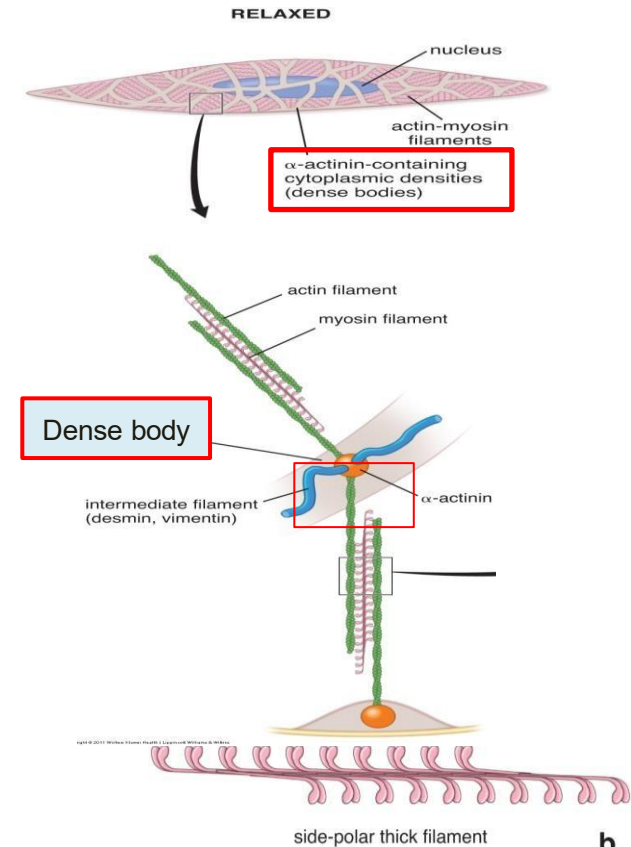
- **Myofilaments**
 - Actin filaments
 - Myosin filaments – smooth muscle myosin (SMM)
- **Sarcoplasmic reticulum (SER)**
 - Sparse profiles unlike striated muscles
- **Caveolae**
 - Invaginations of the sarcolemma
 - **Resembles T-tubules** of striated muscles
 - Lies close to sarcoplasmic reticulum
 - Surrounded by **cytoplasmic vesicles**
 - **Vesicles resemble terminal cisternae, but does not house calcium ions**
- **Calcium (Ca^{2+}) ions**
- **Dense bodies**
 - Analogs of Z-line
 - Distributed throughout the sarcoplasm (cytoplasmic dense bodies) in a network of intermediate filaments or attached to the sarcolemma (plasma membrane dense body)
- **Type III intermediate filaments (desmin, vimentin)**
 - Provide link between dense bodies, cytoskeleton and sarcolemma
 - This link aids in contraction



Associated Proteins of Smooth Muscle Cells' Contractile Apparatus

RP Pg 337

- Essential in the initiation and regulation of contraction
- They include:
 - Myosin light chain kinase (MLCK)
 - initiates contraction
 - Calmodulin
 - Ca^{2+} binding protein
 - Regulates the intracellular calcium concentration
 - Ca^{2+} -calmodulin complex binds & activates MLCK
 - Related to TnC found in skeletal muscles
 - α -Actinin
 - Structural component of dense bodies



How does it work?

In relaxed state

- Free calcium ions are sequestered in the sarcoplasmic reticulum

Nerve stimulation:

- Free calcium is released into the sarcoplasm
 - Binds to calmodulin
 - Calcium-calmodulin complex then activates MLCK
 - Phosphorylates smooth muscle myosin (SMM)
 - SMM, once phosphorylated, can now binds to actin
 - Actin and myosin filaments slide past each other like as in striated muscles
- Because the myofilaments are arranged in a criss-cross pattern inserted around the sarcolemma, contraction results in shortening of the muscle fiber
 - Muscle fiber assumes a globular shape in contrast to the elongated shape in the relaxed state

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Histology of Muscle Tissue - Flipped Class Pre-Lecture Package (3rd Recording – Skeletal Muscle)

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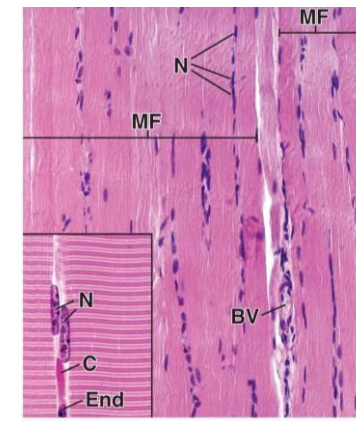
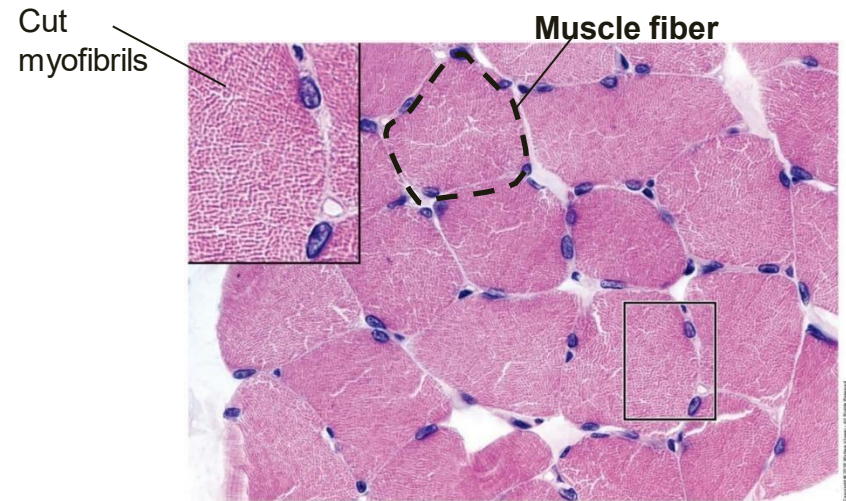
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Skeletal Striated Muscle Tissue

- Usually attached to the skeletal system
- Under voluntary control
- Formed from the fusion of individual myoblasts (muscle cells)

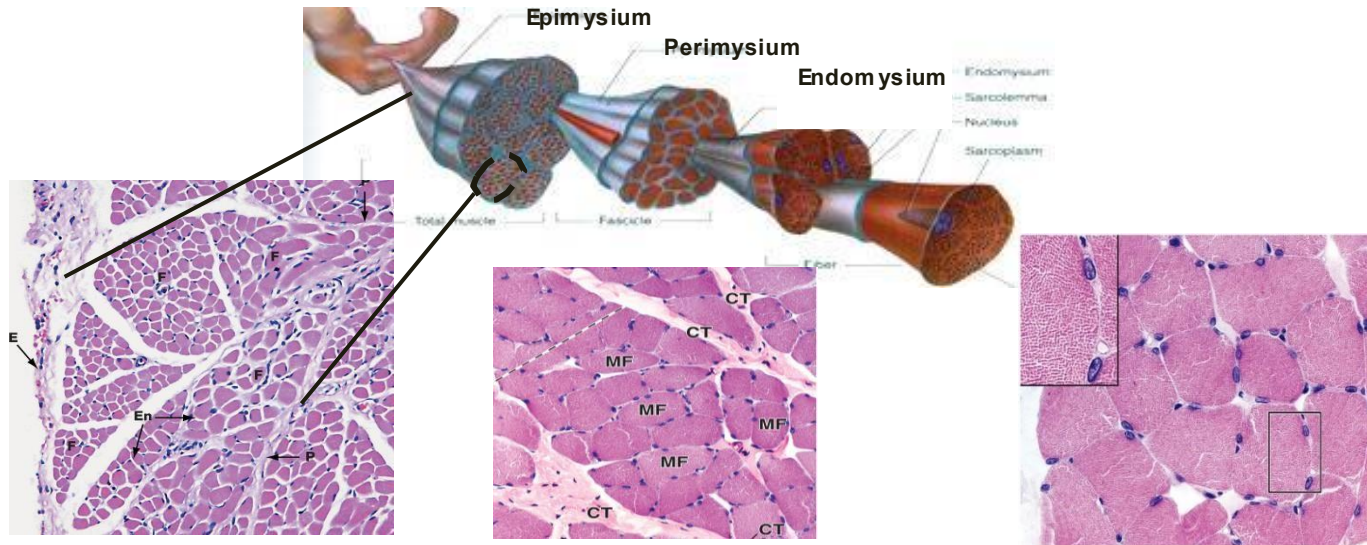
General organization:

- Multinucleated with nuclei just beneath the sarcolemma
- Longitudinal sections:
 - Cylindrical
 - Uniformed length
 - Cross-sections:
 - Regular shaped
 - All cell appear roughly around the same size
- Held together by connective tissue – see next slide



Skeletal Striated Muscle Tissue

Connective tissue coverings



Epimysium- E

Dense connective tissue encasing **multiple** fascicles.

- Contains major blood vessels and nerves
- Continues with tendon to attach muscle at the *myotendinous junction*
- Absent in cardiac and smooth muscles

Perimysium- P

Groups of skeletal myocytes/fibers form a **fascicle**

- Each fascicle is surrounded by a layer of connective tissue or perimysium
- Contains larger blood vessels & Nerves
- Absent in cardiac and smooth muscles

Endomysium- En

Delicate layer of reticular fibers that surrounds **individual** muscle fiber (myocyte)

- Contains small blood vessels and very fine neuronal branches
- Found in cardiac and smooth muscles also

Types of Skeletal Muscles

- **Three types**
 - **Distinguished based on oxidative enzymatic activity**
 - **Color changes (red, white or intermediate) observed when treated with oxidative enzymes such as succinic dehydrogenase, NADH-TR (nicotinamide adenine dinucleotide-tetrazolium reductase) or myosin ATPase**
- **Type I/slow oxidative/red**
 - Small
 - Appears red in fresh specimen
 - Has many mitochondria
 - Reacts strongly with succinic dehydrogenase and NADH-TR histochemical staining
 - Has large amounts of myoglobin and cytochrome complexes
 - Myoglobin is an oxygen binding protein with a ferrous form of iron (Fe^{2+})
 - Slow twitch, fatigue resistant
 - Common in marathon runners
- **Function**
 - Oxygen storage in muscle cells and provide a ready source for muscle metabolism
- **Location: limb muscles**

Types of Skeletal Muscles

- **Type II A/fast oxidative glycolytic/intermediate**
 - Medium sized
 - Appears intermediate in fresh specimen
 - Has high myosin ATPase activity with histochemical staining
 - Has high myoglobin content and abundance of glycogen
 - Capable of anerobic glycolysis
 - Fast twitch, fatigue resistant
 - Common in 400 – 800 m sprinters
- **Type II B/fast glycolytic**
 - Large
 - Appears light pink in fresh specimen
 - Has less myoglobin
 - Has less mitochondria
 - Low level of oxidative enzymes but high anaerobic enzyme activity
 - Produce a lot of lactic acid
 - Fatigue easily
 - Stores a large amount of glycogen
 - Fast twitch, fatigue prone
 - Location: extraocular muscles, digits
 - Common in field athletes, weight-lifters, short sprinters

“Quiz Yourself”

A 16-year-old boy is brought to the physician because of a 1-day history of dark colored urine, and a 4-day history of muscle pains and fatigue. He recently completed a marathon 8 days ago. Physical examination shows myedema (tissue swelling) of both calf muscles. Laboratory studies show mildly deranged kidney function, electrolyte abnormalities and elevated muscle enzymes. Which of the following muscle types is most likely affected in this case?

- A. Type II B**
- B. Type II A**
- C. Type I**

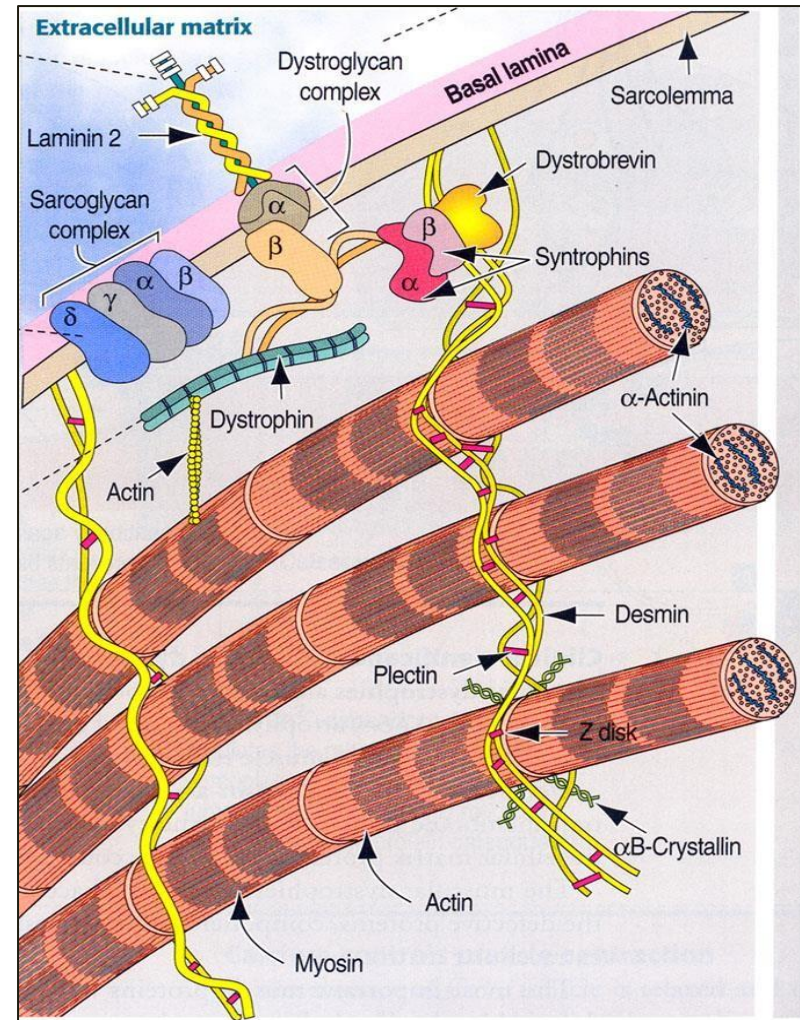
“Quiz Yourself”

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- A. Intense staining reaction**
- B. Poor staining reaction**

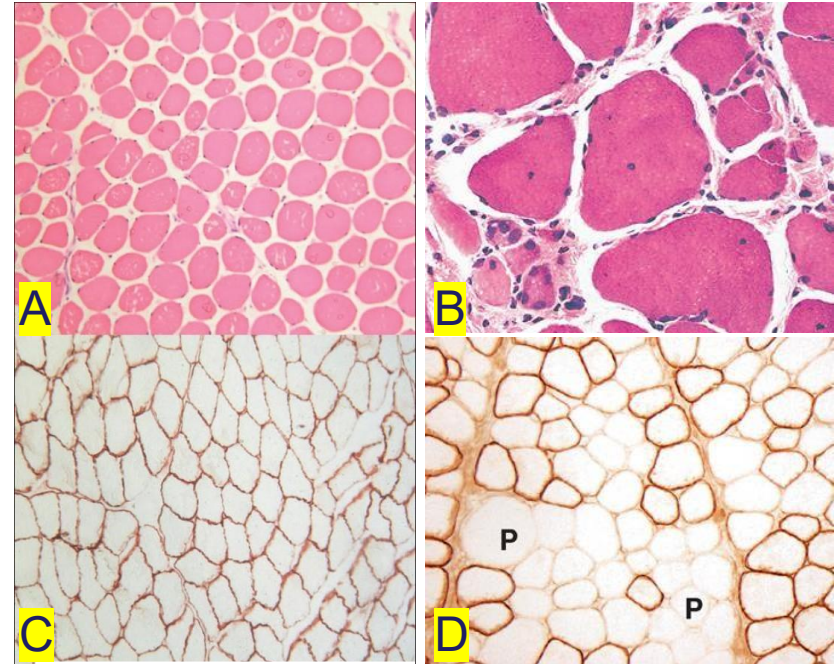
Accessory Proteins of Skeletal Muscles

- **Dystrophin and Dystrophin-Associated Proteins**
- **Dystrophin**
 - A rod-shaped cytoskeletal protein
 - Located just beneath the sarcolemma
 - Has a short head and a long tail
 - F-actin is bound at the end portion of the tail
 - Links the muscle fiber to ECM proteins laminin
 - Dystrophin forms a complex with two groups of transmembrane proteins:
 - Dystroglycans
 - links dystrophin and laminin of the ECM
 - Sarcoglycans
 - associated with membrane dystroglycans



Duchenne Muscular Dystrophy

- DMD results from a defect in the gene coding for dystrophin associated proteins on X chromosome → muscle fiber fragility
- Most common inherited myopathy
- More prevalent in males
- X-linked recessive
- Patients are unable to stand unaided in early childhood and develop progressive muscle weakness, becoming wheelchair-bound by their mid-teens and typically dying in early adult life



Normal skeletal muscle

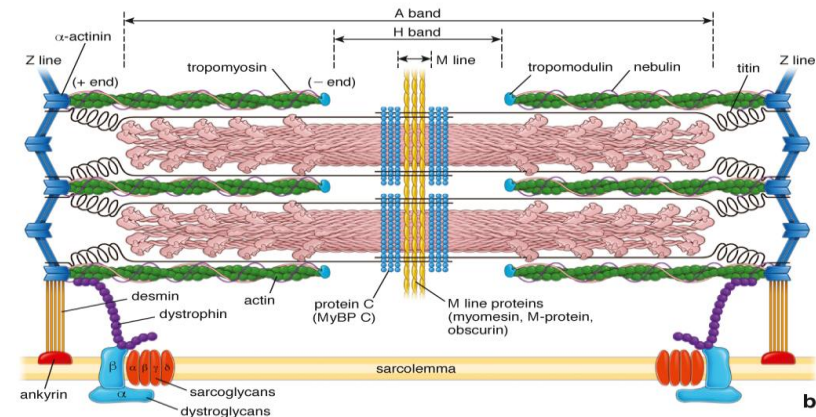
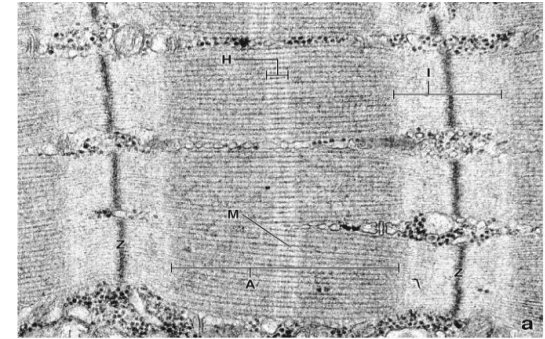
Duchenne Muscular dystrophy

In B there is breakdown of muscle fibers which are then removed by phagocytosis.

In D muscle fibers without dystrophin are seen (P)

Other Accessory Proteins of Skeletal Muscles Cells

- Essential in regulating the spacing, attachment and alignment of the myofilaments within the sarcomere
- Includes:
- **Titin**
 - Extends from the Z-line toward the M line
 - “Molecular springs” – prevents excessive stretching of the sarcomere
 - Helps centre the thick filaments between the Z-lines
- **Desmin**
 - Intermediate filament
 - Attaches Z-line to each other and to the plasma membrane
 - Uses linker protein ankyrin



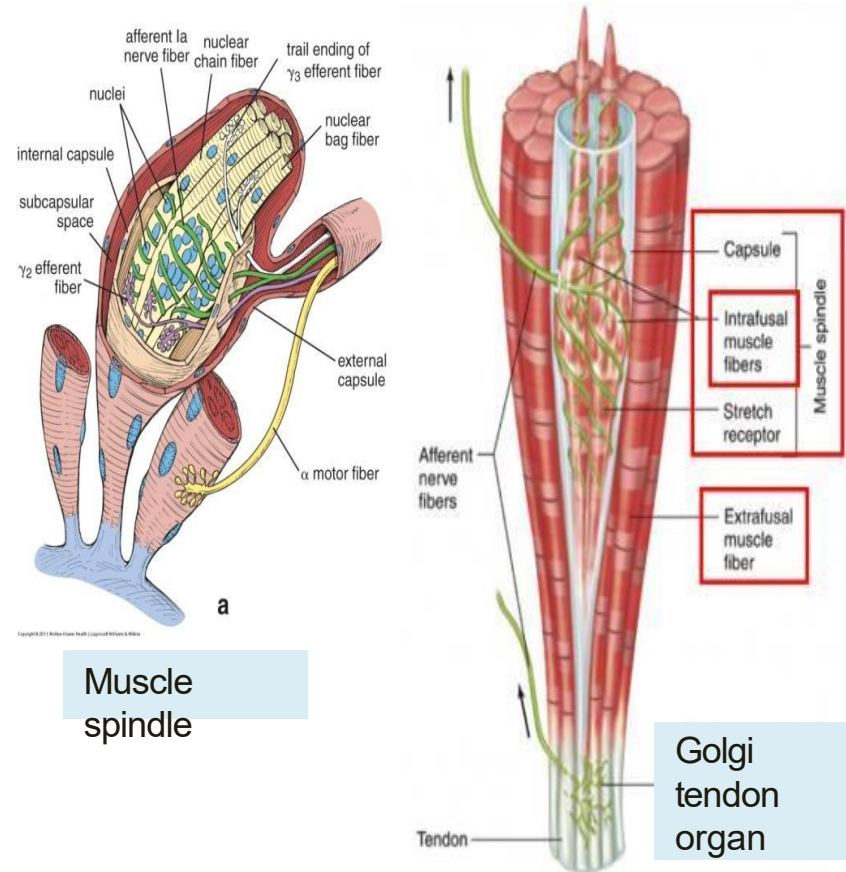
Sensory Innervation

Muscle Spindle

- Encapsulated sensory receptors
- Specialized stretch receptor located in the muscle belly
- Senses changes in muscle length or stretch
- Contains modified muscle fibers called Spindle Cells and neuronal terminals

Golgi tendon organ

- Encapsulated proprioceptor located at the myotendinous junction
- Senses tension in the muscle
- Sensory component of the Golgi tendon reflex/deep tendon reflex



Basic Principles of Medicine 1

Module: Foundations To Medicine| Lecture 35

Histology of Muscle Tissue - Flipped Class Pre-Lecture Package (4th Recording – Cardiac Muscle)

Dr. Rachael George

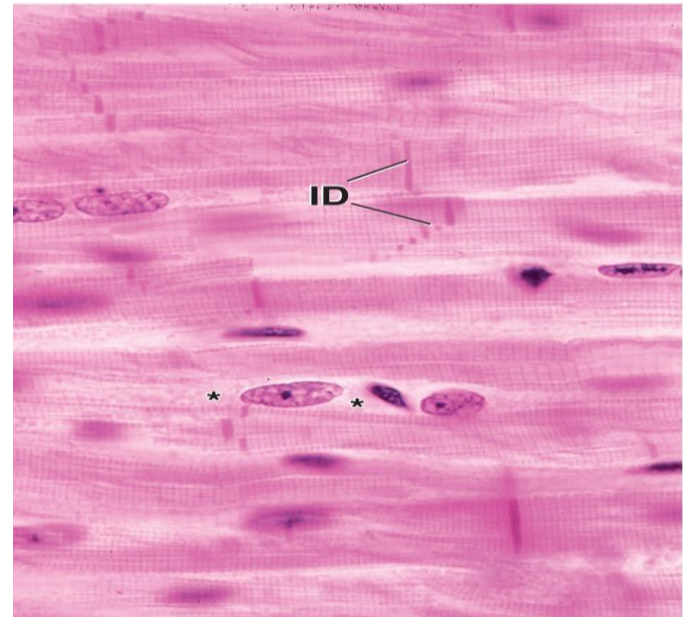
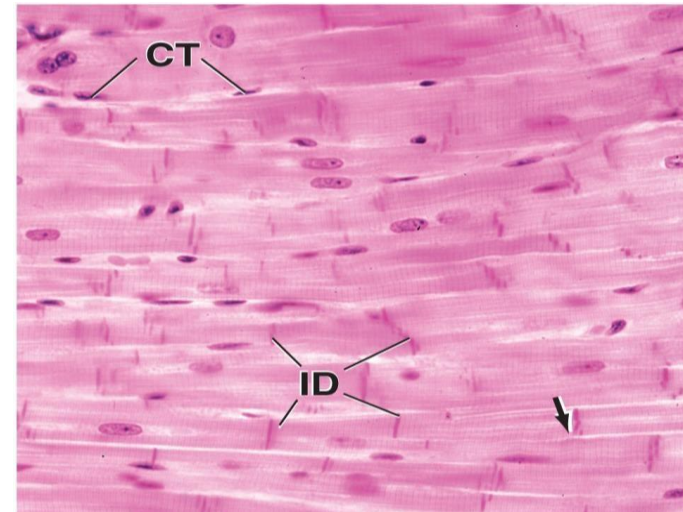
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School of Medicine, St. George's University

Cardiac Striated Muscle Tissue

- Forms the bulk of the myocardium
- May extend slightly into the walls of some of the great vessels
- Involuntary control
- Muscle fibers (cardiac myocytes)
 - Branched
 - Surrounded by endomysium
 - Single, centrally located nuclei
 - Some may be bi-nucleated
 - Cylindrical in long axis, but is of varying lengths
 - Arranged end-to-end
 - Held in place by intercalated discs
- Perinuclear region (*) is free of myofibrils and houses the organelles
- Numerous large mitochondria & glycogen stores
- Granules containing diuretic hormones: atrial natriuretic peptide and brain natriuretic peptide

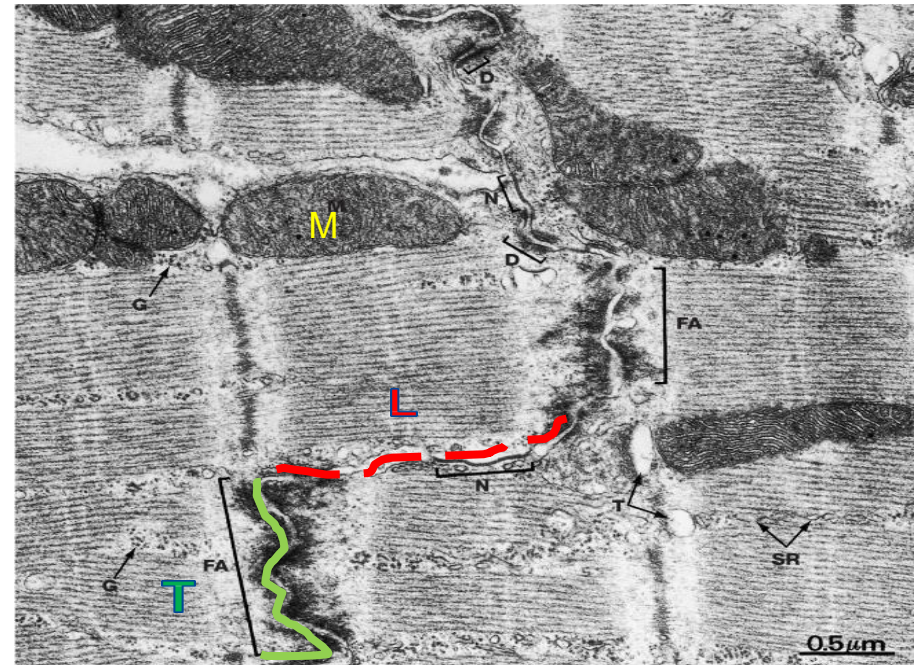
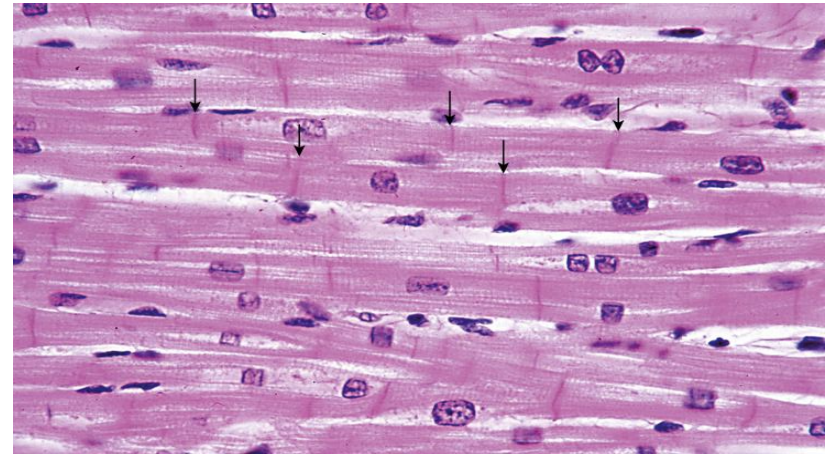


ID – intercalated discs

* - perinuclear region

Cardiac Muscle: Intercalated disc

- Intercalated discs are attachment sites between adjacent cardiac myocytes
- **Transversely (T)** oriented parts of the intercalated disc
 - seen at right angle to the myofibril like the risers of a stairway
 - Contains macula adherens and fascia adherens
- **Longitudinal or lateral parts (L)** are parallel to myofibril like the steps of a stairway
 - Contains gap junctions and macula adherens
- **Mitochondria (M)** are abundant in cardiac muscle due to the high metabolic demands of these cells



Cardiac Muscle: Intercalated disc

Transverse component (T)

(FA) Fascia adherens (adhering junctions)

- Major structural element
- Binds cardiac muscle cells at their ends
- Serves as attachment site for thin filaments in terminal sarcomeres

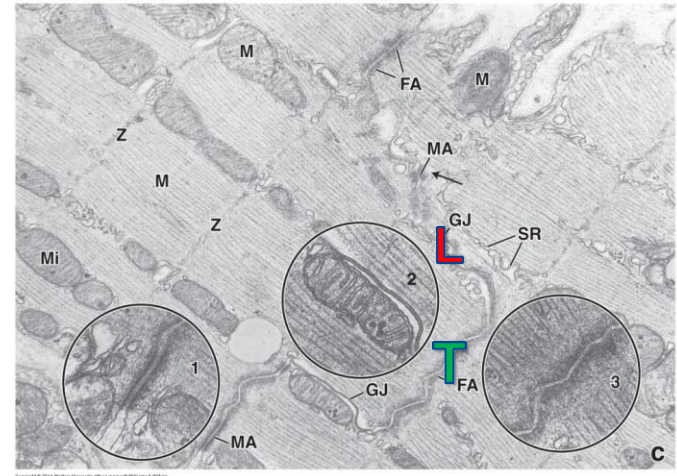
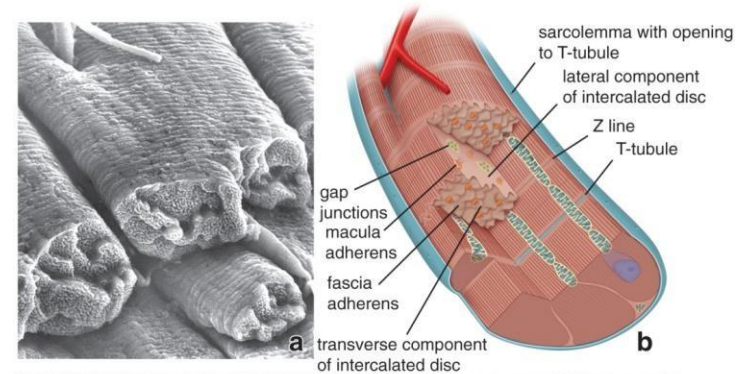
(MA) Maculae adherentes (desmosomes)

- Bind individual muscle cells to each other
- Reinforce fascia adherens
- Found in both transverse & lateral component.

Lateral component (L)

(GJ) Gap junctions (communicating junctions)

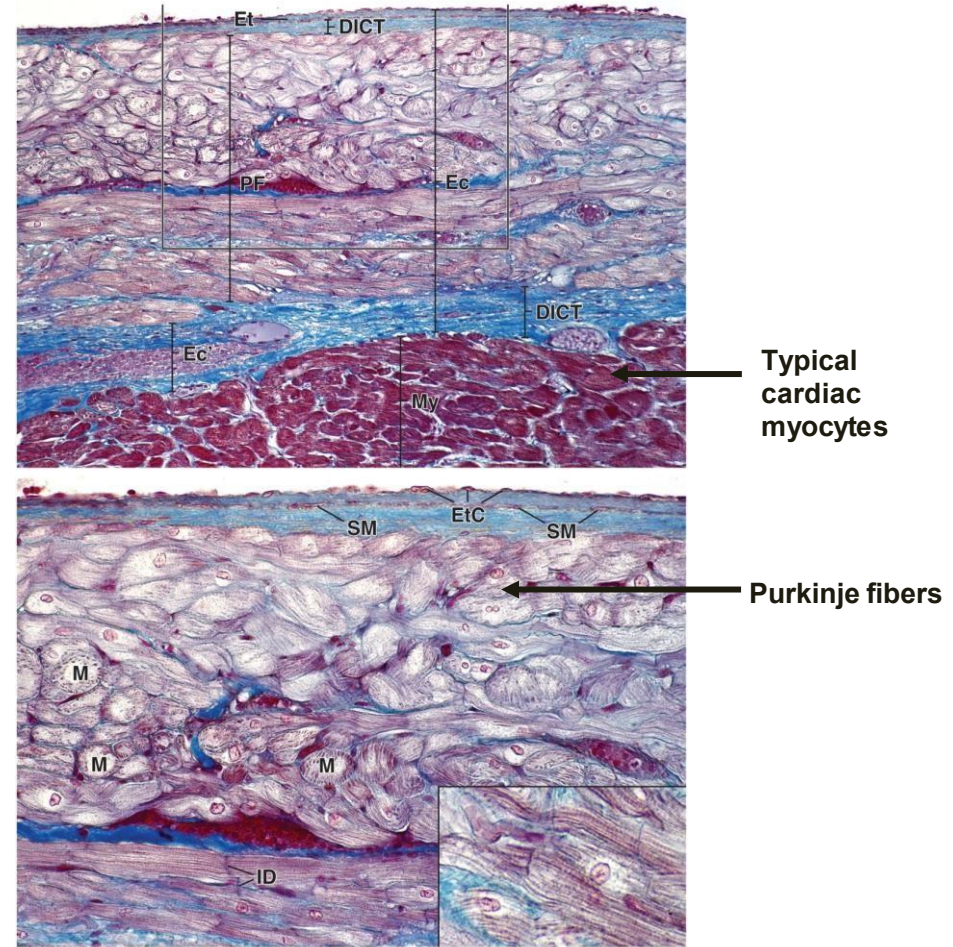
Maculae adherents (MA)



Cardiac Muscle

Purkinje fibers

- Large, modified cardiac muscle cells located just deep to the endocardium in the **subendocardial** connective tissue
- Specialized to conduct impulses of the A-V bundle and allow synchronization of ventricular contraction
- Pale staining due to
 - few myofibrils(these are located peripherally)
 - large amount of glycogen
- Abundant mitochondria





Basic Principles of Medicine 1

Module: Foundations To Medicine| Lecture 35

Histology of Muscle Tissue - Flipped Class Pre-Lecture Package (5th Recording – Smooth Muscle)

Dr. Rachael George

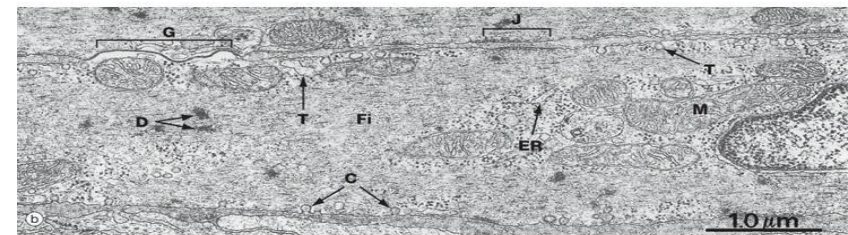
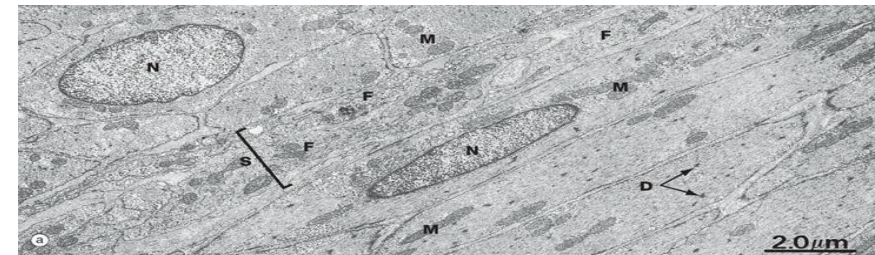
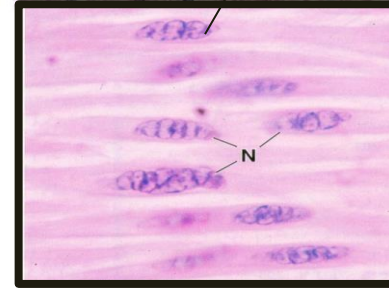
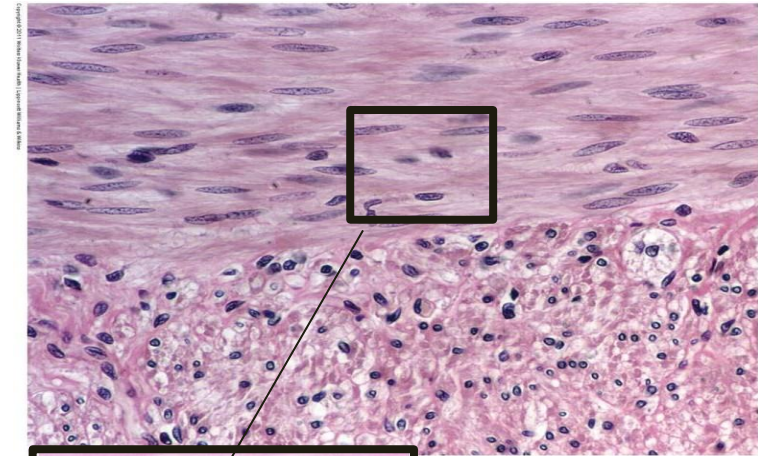
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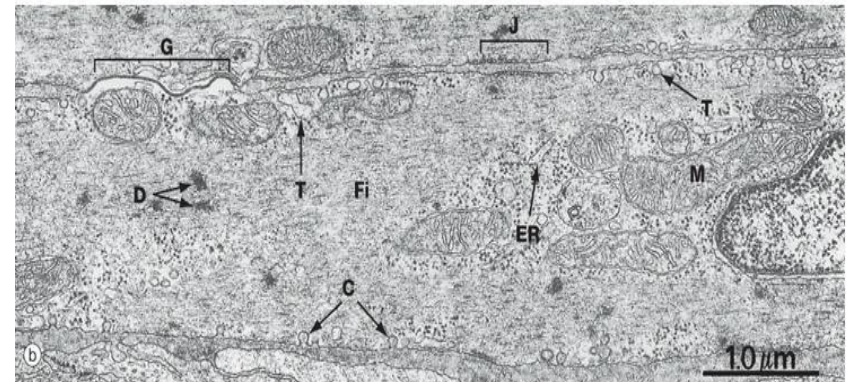
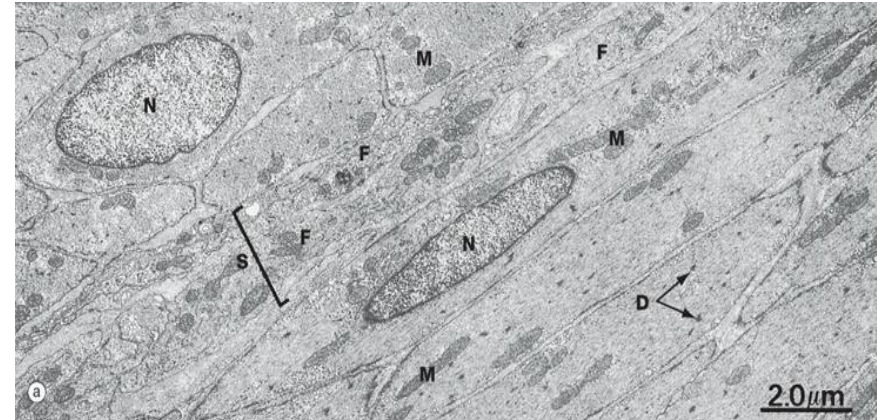
Smooth Muscle

- Elongated, fusiform cells with tapered ends
- Interconnected via gap junctions ((J) in EM below)
- Generally organized into bundles or sheets around the lumen of hollow organs e.g. gastrointestinal tract, uterus, blood vessels
- Central nucleus
 - Appears cork-screwed in contracted state
 - Appears rounded in cross section
- No cross-striations thus even staining with H&E staining
 - Lacks a sarcomere
 - Still possess contractile proteins
- Sarcoplasm
 - Pinocytotic vesicles
 - Dense bodies (D)
 - Forms attachment sites for thin (actin) filaments and intermediate filament (desmin or vimentin)
 - Composed of plaque proteins eg α -actinin



Smooth Muscle

- **Smooth muscle (SM) cells secrete connective tissue matrix**
- Synthesize both type IV collagen and type III collagen
- Elastin, proteoglycans, and multi-adhesive glycoproteins
- Vascular and uterine SM also secrete large amounts of type I collagen and elastin
- **Possess:**
 - Well-developed rER (ER) & golgi (G)



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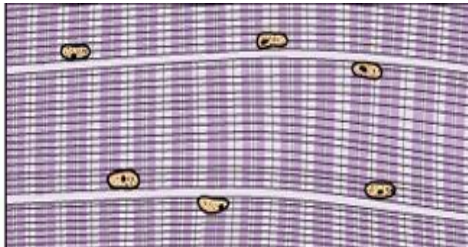
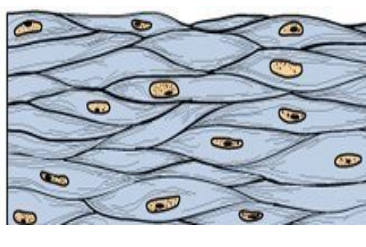
S – Sarcoplasmic reticulum
 D – Dense bodies
 M – Mitochondria
 C – Caveolae

Other Proteins of Smooth Muscle Tissue

- Note that these proteins are not associated with either of the two myofilaments
- Include:
 - Calmodulin
 - Ca^{2+} binding protein
 - Regulates the intracellular calcium concentration
 - Ca^{2+} calmodulin complex binds & activates Myosin Light Chain Kinase (MLCK)
 - Related to TnC found in skeletal muscles
 - Intermediate filaments
 - Desmin, vimentin
 - Provide the link between the dense bodies, the cell's cytoskeleton and the sarcolemma
 - Allows for stabilization of the dense bodies during contraction as well as allowing the inward movement of the sarcolemma

	Skeletal	Cardiac	Smooth
Hypertrophy	Yes	Hypertrophy as seen in hypertensive cardiomyopathy	Yes
Hyperplasia	No	No	Yes SM of the uterus proliferates during the normal menstrual cycle and pregnancy SM of blood vessels increase in size as well as number due to hypertensive conditions etc
Response to injury	Limited capacity to regenerate Satellite cells are myogenic precursor cells located between the sarcolemma of a muscle fiber and its external lamina Satellite cells are responsible for skeletal muscle regeneration	No regenerative capacity After the death of cardiac muscle cells, the tissue is replaced with fibrous connective tissue. Cardiac function is lost at the site of injury.	SM of blood vessels divide regularly differentiating from mesenchymal stem cells in the adventitia or from division and differentiation of endothelial cells and pericytes in capillaries and post-capillary venules during the repair process following a vascular injury.

Summary Table

Classification	Striated muscle These tissue subtypes have visible cross striations due to regular arrangement of contractile organelles within their cells			Smooth muscle No cross striations
Type	Skeletal	Visceral	Cardiac	
Contraction	Voluntary	Voluntary	Involuntary	Involuntary
Location	Somatic/ body wall	Soft tissue origin. Tongue, pharynx, larynx, diaphragm and upper esophagus	Heart and roots of great veins that empty into the heart	Walls of visceral organs. Stomach, gut tube etc.
General morphology				

Summary Table

	Skeletal	Cardiac	Smooth
			
Structural features			
Muscle cell	Large, elongate cell, 10–100 μm in diameter, up to 100 cm in length (sartorius muscle)	Short, narrow cell, 10–100 μm in diameter, 80–100 μm in length	Short, elongate, fusiform cell, 0.2–2 μm in diameter, 20–200 μm in length
Location	Muscles of skeleton, visceral striated (e.g., tongue, esophagus, diaphragm)	Heart, superior and inferior vena cava, pulmonary veins	Vessels, organs, and viscera
Structural features			
Connective tissue components	Epimysium, perimysium, endomysium	Endomysium, subendocardial and subpericardial connective tissue	Endomysium, sheaths, and bundles
Fiber	Single skeletal muscle cell	Linear branched arrangement of several cardiac muscle cells	Single smooth muscle cell
Sarcomere	Present	Present	None
Striation	Present	Present	None
Nucleus	Many peripheral	Single central, surrounded by juxtanuclear region	Single central
T tubules	Present at A–I junction (triad: with two terminal cisternae), two T tubules/sarcomere	Present at Z lines (diad: with small terminal cisternae), one T tubule/sarcomere; Purkinje fibers have less number of T tubules.	None, well-developed sER, many invaginations and vesicles similar to caveolae
Cell-to-cell junctions	None	Intercalated discs containing 1. Fasciae adherentes 2. Macula adherens (desmosome) 3. Gap junctions	Gap junctions (nexus)
Special features	Well-developed sER and T tubules	Intercalated discs, Purkinje fibers	Dense bodies, desmin and vimentin filaments, caveolae, and cytoplasmic vesicles
Functions			
Type of innervation	Voluntary	Involuntary	Involuntary
Efferent innervation	Somatic	Autonomic	Autonomic
Type of contraction	“All or none” (type I and type II fibers)	“All or none” rhythmic (pacemakers, conductive system of the heart)	Slow, partial, rhythmic, spontaneous contractions (pacemakers of stomach)
Regulation of contraction	By binding of Ca^{2+} to TnC, causes tropomyosin movement and exposes myosin-binding sites on actin filaments	By binding of Ca^{2+} to TnC, causes tropomyosin movement and exposes myosin-binding sites on actin filaments	By phosphorylation of myosin light chain by myosin light chain kinase in the presence of Ca^{2+} –calmodulin complex
Growth and regeneration			
Mitosis	None	None (in normal condition)	Present
Response to demand	Hypertrophy	Hypertrophy	Hypertrophy and hyperplasia
Regeneration	Limited (satellite cells and myogenic cells from bone marrow)	None (in normal condition)	Present

sER, smooth-surfaced endoplasmic reticulum; TnC, troponin-C.